MATHEMATICS

SECTION – I

Single Correct Choice Type

1. **(a)** $f(x)=\cos x\int_0^x \sin(t^2-t)dt+\sin x\int_0^x \cos(t^2-t)dt$

$f''(x)=-\sin x\sin(x^2-x)+(2x-1)\cos x\cos(x^2-x)$

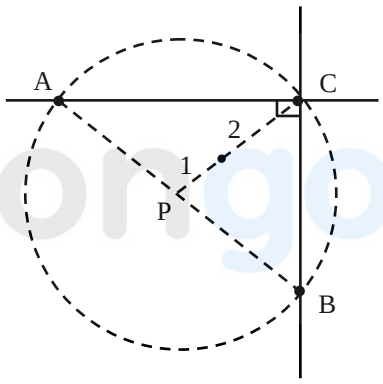
$-\sin x\sin(x^2-x)-\cos x\int_0^x \sin(t^2-t)\,dt+\cos x\cos(x^2-x)$

$+(1-2x)\sin x\sin(x^2-x)-\sin x\int_0^x \sin(t^2-t)dt+\cos x\sin(x^2-x),$

Then $f(x)+f''(x)=(2x+1)\cos x^2$

$f\left(-\frac{1}{2}\right)+f''\left(-\frac{1}{2}\right)=0$

2. **(a)** As lines are perpendicular to each other 'C' moves on a circle with AB as diameter.
Now P, mid-point of AB (which is fixed) when joined with C is median.
$\Rightarrow$ Centroid is moving at a constant distance $\frac{1}{3}(PA)$ from P.
$\Rightarrow$ Locus is a circle



A is point of intersection of $x+4y+2=0$ and $x-y+1=0$ i.e., $\left(-\frac{6}{5},-\frac{1}{5}\right)$

B is point of intersection of $4x-y+6=0$ and $x+y+3=0$ i.e., $\left(\frac{-9}{5},-\frac{6}{5}\right)$

$\Rightarrow P=\left(\frac{-3}{2},\frac{-7}{10}\right)$ therefore locus is $\left(x+\frac{3}{2}\right)^2+\left(y+\frac{7}{10}\right)^2=\frac{34}{225}$

3. **(b)** Consider $a<b$, then $|f(b)-f(a)|<b-a$ …(i)

Also $f(0)=f(1)$, therefore

$|f(b)-f(a)|\le|f(b)-f(0)|+|f(1)-f(a)|$ …(ii)

$|f(b)-f(0)|+|f(1)-f(a)|<1-a+b$ ….(iii)

Adding (i), (ii) and (iii), we get

$|f(b)-f(a)|\le\frac{1}{2}$

4. **(c)** $84=2^2\times3\times7$

$496! = 2^{\lambda}\, 3^{\beta}\, 5^{\gamma}\, 7^{\delta} \ldots\ldots$

Here $\beta > \delta$ and $\lambda > \delta$ as the frequency of occur of 2 is many a times more than that of 3 and 7.

Therefore divisibility will be decided by δ and $\delta = \left[\frac{496}{7}\right] + \left[\frac{496}{49}\right] + \ldots.$ (where [.] greatest integer function)

$= 70 + 10 = 80$

5. **(b)** By inspection $\frac{2\pi}{3}$ is the fundamental period.

Alternately:

If $z = \cos x + i \sin x$; $\frac{1}{z} = \cos x - i \sin x$, we get $128\cos^7 x = \left(z + \frac{1}{z}\right)^7$

$\Rightarrow \quad 64\cos^7 = \cos 7x + 7\cos 5x + 21\cos 3x + 35\cos x$

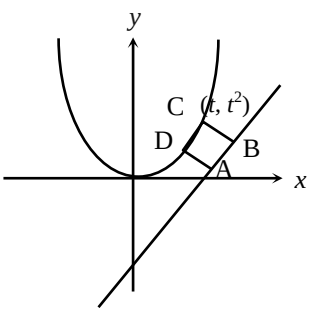
Adding after putting x as $\left(x + \frac{2\pi}{3}\right)$ and $\left(x + \frac{2\pi}{3}\right)$

$64\left(\cos^7 x + \cos^7\left(x + \frac{2\pi}{3}\right) + \cos^7\left(x + \frac{4\pi}{3}\right)\right)$, we get $63\cos 3x$

$\cos^7 x + \cos^7\left(x + \frac{2\pi}{3}\right) + \cos^7\left(x + \frac{4\pi}{3}\right) = \frac{63}{64}\cos(3x)$

Hence period is $\frac{2\pi}{3}$

6. **(a)** $BC = \frac{\left|2t - t^2 - 17\right|}{\sqrt{5}} = a$



$\Rightarrow \quad t^2 - 2t + 17 - \sqrt{5}a = 0 \quad \ldots(i)$

If $C(t_1, t_1^2)$, $D(t_2, t_2^2)$, then

t_1, t_2 are roots of equation (i) and $CD = a$

$\Rightarrow \quad t_1 + t_2 = 2, \quad t_1 t_2 = 17 - \sqrt{5}a$

$\Rightarrow \quad a^2 = (t_1 - t_2)^2 + (t_1^2 - t_2^2)^2$

$a^2 = 4 - 4(17 - \sqrt{5}a) + 4(4 - 4(17 - \sqrt{5}a))$

$a^2 = 20 - 20(17) + 20\sqrt{5}a$

$\Rightarrow \quad a^2 - 20\sqrt{5}a + 320 = 0 \quad \Rightarrow \quad a = 16\sqrt{5}$ or $a = 4\sqrt{5}$

Minimum area is when $a = 4\sqrt{5} \quad \Rightarrow \quad a^2 = 80$

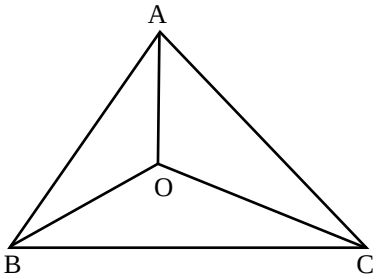
7. **(a)** The vector equation of the plane passing through points $\vec{a}, \vec{b}, \vec{c}$ is

$r.\left(\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right)=[\vec{a}\ \vec{b}\ \vec{c}]$

Therefore, the length of the perpendicular from the origin to this plane is given by

$$\frac{[\vec{a}\ \vec{b}\ \vec{c}]}{|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}|}.$$

8. **(c)**



$$\frac{\text{area of }(\Delta ABC)}{\text{area of }(\Delta OAC)}$$

$$=\frac{\frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OB}\right|+\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|+\left|\overrightarrow{OC}\times\overrightarrow{OA}\right|}{\frac{1}{2}\left|\overrightarrow{OA}\times\overrightarrow{OC}\right|}$$

$$=\frac{\left|\overrightarrow{OA}\times\overrightarrow{OB}\right|+\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|}{\left|\overrightarrow{OA}\times\overrightarrow{OC}\right|}+1$$

Given $\overrightarrow{OA}+2\overrightarrow{OB}+3\overrightarrow{OC}=\vec{0}$

$2\overrightarrow{OB}\times\overrightarrow{OA}+3\overrightarrow{OC}\times\overrightarrow{OA}=\vec{0}$

$$\Rightarrow \frac{\left|\overrightarrow{OB}\times\overrightarrow{OA}\right|}{3}=\frac{\left|\overrightarrow{OC}\times\overrightarrow{OA}\right|}{2}=\lambda \qquad \ldots(i)$$

$\overrightarrow{OB}\times\overrightarrow{OA}+3\overrightarrow{OB}\times\overrightarrow{OC}=\vec{0}$

$$\Rightarrow \frac{\left|\overrightarrow{OB}\times\overrightarrow{OA}\right|}{3}=\frac{\left|\overrightarrow{OB}\times\overrightarrow{OC}\right|}{1}=\lambda \qquad \ldots(ii)$$

$\Rightarrow$ Ratio of area of Δs i.e., $\frac{\Delta ABC}{\Delta AOC}=\frac{3\lambda+\lambda}{2\lambda}+1=3$

9. **(d)** Total five digit numbers are 90000.

Every third term is divisible by 3 therefore numbers divisible by 3 are 30000.

The multiples of 3 end periodically with digits 3, 6, 9, 2, 5, 8, 1, 4, 7 and 0.

Hence numbers that end with digit 6 are 3000.

Alternately:

Let *abcd* 6 be the five digit numbers.

Therefore the four digit numbers '*abcd*' should be divisible by 3 i.e., $9000\div 3$ i.e, 3000.

SECTION – II
Multiple Correct Choice Type

10. (a), (c)

Let the drawer contains p balls of which 'm' are red.

Probability of drawing two red balls at random is $\frac{{}^{m}C_2}{{}^{p}C_2}=\frac{1}{2}$

$\Rightarrow \quad 2m(m-1)=p(p-1)$

$\Rightarrow \quad 2m^2-2m-p^2+p=0$

$\Rightarrow \quad m=\frac{2\pm\sqrt{4-8(p-p^2)}}{4}=\frac{1\pm\sqrt{1-2p+2p^2}}{2}$

$\Rightarrow \quad 1-2p+2p^2$ should be an odd perfect square.

i.e., $p=21,\ 4$ but $p\neq 3$

when 3 balls out of 4 are red.

15 balls out of 21 are red.

11. (a), (b), (c)

$(x-y)f(x+y)-(x+y)f(x-y)=2y(x-y(x+y))$

Let $x-y=u;\ x+y=v$

$uf(v)-vf(u)=2uv(v-u)$

$\frac{f(v)}{v}-\frac{f(u)}{u}=v-u$

$\left(\frac{f(v)}{v}-v\right)=\left(\frac{f(u)}{u}-u\right)$=constant

Let $\frac{f(x)}{x}-x=\lambda$

$\Rightarrow \quad f(x)=(\lambda x+x^2)$

$f(1)=2$

$\lambda+1=2 \quad \Rightarrow \quad \lambda=1$

$f(x)=x^2+x$

12. (a), (d)

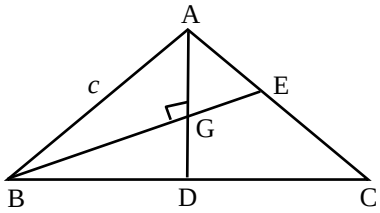
Since $f''(x)>0$ graph of $f(x)$ is concave upward.

Now $f'(x)=\pm\frac{7}{3}$ having a solution would mean line having slope $\pm\frac{7}{3}$ touching the curve.

The graph will strictly lie in the rectangle formed by $x=1,\ x=4,\ y=7,\ y=14$.

If f is monotonic the line parallel to either of diagonal of rectangle will touch the graph exactly once and if f is non-monotonic, then lines parallel to diagonals will touch the graph exactly once.

13. (a), (d)



$AG = \frac{1}{3}\sqrt{2b^2 + 2c^2 - a^2}$ and

$BG = \frac{1}{3}\sqrt{2a^2 + 2c^2 - b^2}$

$AG^2 + BG^2 = c^2$

$\Rightarrow \quad (2b^2 + 2c^2 - a^2) + (2a^2 + 2c^2 - b^2) = 9c^2$

$\Rightarrow \quad a^2 + b^2 = 5c^2 \quad$ ...(i)

and for ΔABC exist

$a - b < c$ or $b - a < c$

$\Rightarrow (a-b)^2 < c^2 \Rightarrow a^2 + b^2 - 2ab < \frac{a^2 + b^2}{5} \Rightarrow 2a^2 + 2b^2 - 5ab < 0$

$\Rightarrow 2\left(\frac{a}{b}\right)^2 - 5\left(\frac{a}{b}\right) + 2 < 0 \Rightarrow \frac{a}{b} = \frac{5 \pm \sqrt{25-16}}{4} = \frac{5 \pm 3}{4} = 2, \frac{1}{2}$

$\Rightarrow \frac{1}{2} < \frac{a}{b} < 2$

Similarly, $\frac{1}{2} < \frac{b}{a} < 2$

SECTION – III
Reasoning Type

14. (d)

15. (c) $a_{ij} = i^2 - j^2 \Rightarrow a_{ji} = j^2 - i^2 = -a_{ij}$

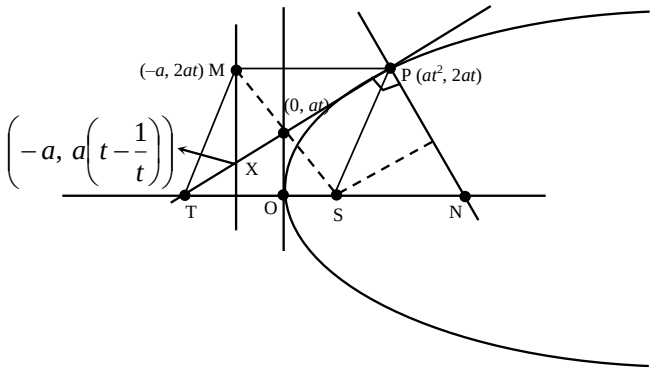
Thus 'A' is necessarily a skew symmetric matrix.

Determinant of skew-symmetric matrix of odd order is always zero.

16. (d)

SECTION–IV
Matrix–Match Type

1. A-1, 3; B-2; C-5; D-4



2. From given equation $\cos^{-1}(y)+\cos^{-1}(bxy)=\frac{\pi}{2}-\sin^{-1}(ax)$

Consider $\cos^{-1} y=\alpha, \cos^{-1}(bxy)=\beta, \cos^{-1}(ax)=\gamma$

$\Rightarrow \alpha+\beta=\gamma \Rightarrow \beta=\gamma-\alpha$ also $\cos\alpha = y, \cos\beta = bxy, \cos\gamma = ax$

$\Rightarrow \cos(\gamma-\alpha)=bxy$

$\Rightarrow \cos\alpha\cos\gamma+\sin\gamma\sin\alpha=bxy \Rightarrow (axy-bxy)^2=\sin^2\gamma\sin^2\alpha=(1-\cos^2\gamma)(1-\cos^2\alpha)$

$\Rightarrow (a-b)^2x^2y^2=(1-a^2x^2)(1-y^2)$

(A)- 1, 5 If $a = 1, b = 0$, then $x^2+y^2=1$

(B)- 2 If $a = 1, b = 1$, then $(x^2-1)(y^2-1)=0$

(C)- 1, 5 If $a = 1, b = 2$, then $x^2+y^2=1$

(D)- 3, 4 If $a = 2, b = 0$, then $4x^2+y^2=1$

SECTION–V
Integer Answer Type

1. Since it is an identity the value of L.H.S. and R.H.S. are equal for all values of x

Put $x=0 \Rightarrow \alpha_1\alpha_2\alpha_3\alpha_4=6$

Ans. 6

2. Given inequality can be written as:

$f''(x)-2f'(x)\geq 3(f'(x)-2f(x))$

Let $f'(x)-2f(x)=g(x)$

$\Rightarrow g'(x)-3g(x)\geq 0$ Multiply e^{-3x}

$\Rightarrow \left(g(x)e^{-3x}\right)'\geq 0 \Rightarrow g(x)e^{-3x}$ is non-decreasing.

Now $g(0)=f'(0)-2f(0)=-2$

$g(x)e^{-3x}\geq -2, \forall x\geq 0$

$f'(x)-2f(x)\geq -2e^{3x}, \forall x\geq 0$ Multiply e^{-2x}

$\Rightarrow \left(f(x)e^{-2x}\right)'\geq -2e^{x}, \forall x\geq 0$

$\Rightarrow \left(f(x)e^{-2x}+2e^{x}\right)'\geq 0$

$\Rightarrow f(x)e^{-2x}+2e^{x}\geq 3$

$\Rightarrow f(x)\geq 3e^{2x}-2e^{3x}, \forall x\geq 0$

Comparing $ah(bx)-bh(ax)$ with $3e^{2x}-2e^{3x}$ we get $h(x)=e^x, a=3, b=2$

$\Rightarrow (a+b)h(0)=5$

Ans. 5

3. Since $[\sqrt{2046}=[\sqrt{2047}]=[\sqrt{2048}]=[\sqrt{2049}=45$

$\therefore$ 2003rd term is $2003+45=2048$

Hence remainder is 0

Ans. 0

4. $y=\sqrt{9-a^2+2ax-x^2}$ $\Rightarrow$

$(x-a)^2+y^2=9$

For given inequality to hold for positive x.

$a-3<4$

$\Rightarrow$ $a<7$ $\Rightarrow$ $a=6$

Ans. 6

